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Division: 2 (G3)

Course: BlockChain Technology

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Assignment - 2

Lab-2: for every block, consider difficulty level is 0, 00,000,000. now try to find out the hash of each block according to suitable nonce value. Draw the graph how much take by each case in order to find out the nonce value. Write your observation.

Code:

```
import java.security.MessageDigest;

public class MineBlock {

    // SHA-256 Hash Function
    public static String sha256(String input) throws Exception {

        MessageDigest md = MessageDigest.getInstance("SHA-256");
        byte[] hash = md.digest(input.getBytes());

        StringBuilder sb = new StringBuilder();

        for (byte b : hash) {
            sb.append(String.format("%02x", b));
        }

        return sb.toString();
    }

    public static void mine(String message, int difficulty) throws Exception {

        int nonce = 0;
        String target = "";

        // Create target like "0", "00", "000"
        for (int i = 0; i < difficulty; i++) {
            target += "0";
        }

        long startTime = System.currentTimeMillis();
```

```

while (true) {

    String data = message + nonce;

    String hash = sha256(data);

    if (hash.startsWith(target)) {

        long endTime = System.currentTimeMillis();

        double time = (endTime - startTime) / 1000.0;

        System.out.println("Difficulty : " + difficulty);
        System.out.println("Nonce      : " + nonce);
        System.out.println("Hash       : " + hash);
        System.out.println("Time       : " + time + " seconds");
        System.out.println();

        break;
    }

    nonce++;
}

public static void main(String[] args) throws Exception {

    String message = "23BCP096";

    int[] difficulties = {1, 2, 3};

    for (int difficulty : difficulties) {
        mine(message, difficulty);
    }
}
}

```

Output:

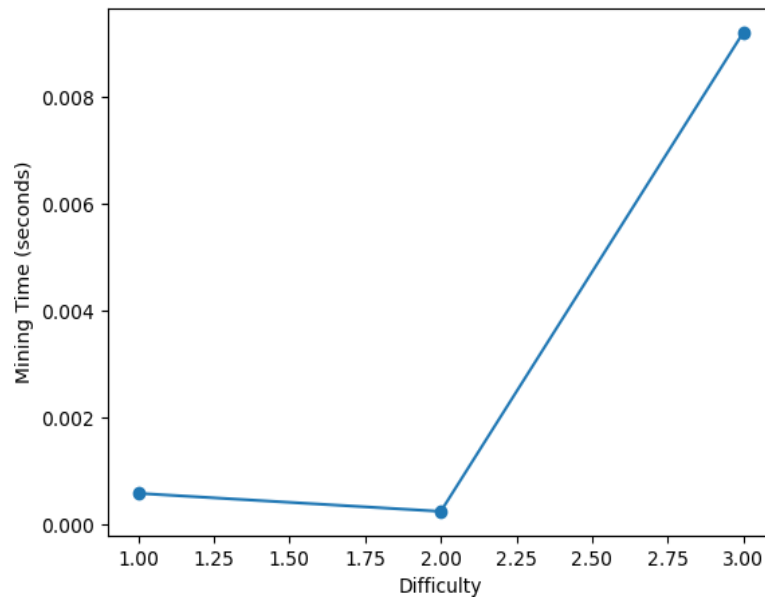
```
[Running] cd "b:\desktop\JAVA\Greedy\" && javac MineBlock.java && java MineBlo
Difficulty : 1
Nonce      : 17
Hash       : 095c7547e093ab3d289d9d7d72feaec4d1a2ca097df4000af31688a71cbf42aa
Time       : 0.041 seconds

Difficulty : 2
Nonce      : 447
Hash       : 008da5cb56faaf07006c11c295dcf02ca376d1f298ec502a528f5f6a2ad9d342
Time       : 0.079 seconds

Difficulty : 3
Nonce      : 5093
Hash       : 0005072fa7e151c0c8f2b5c4ead929481e3a05e5cd1517eb8ed9229ec56a7614
Time       : 0.3 seconds

[Done] exited with code=0 in 1.569 seconds
```

Graph:



Observation:

In this experiment, I observed that the time required to find a valid hash increases as the difficulty level increases. For lower difficulty, the required nonce was found quickly, but for higher difficulty, more attempts were needed. This shows how the Proof of Work mechanism requires more computational effort when the difficulty is increased.